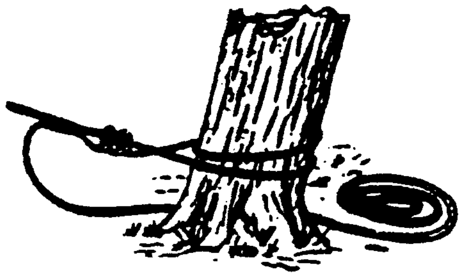
Appendix D
Expedient Anchorages

Holdfasts

Natural

Trees, stumps, or rocks can serve as natural anchorages for rapid work in the field. Always attach lines near ground level on trees or stumps. Avoid dead or rotten trees as they are likely to snap suddenly when strain is placed on the anchor line. It is always advisable to lash the first tree or stump to a second one, to provide added support. A transom can be placed between two trees to provide a sturdier support. When using rocks as anchorages, examine the rocks carefully to be sure that they are large enough and firmly embedded in the ground. An outcropping of rocks or a heavy boulder buried partially in the ground can serve as a satisfactory anchor.

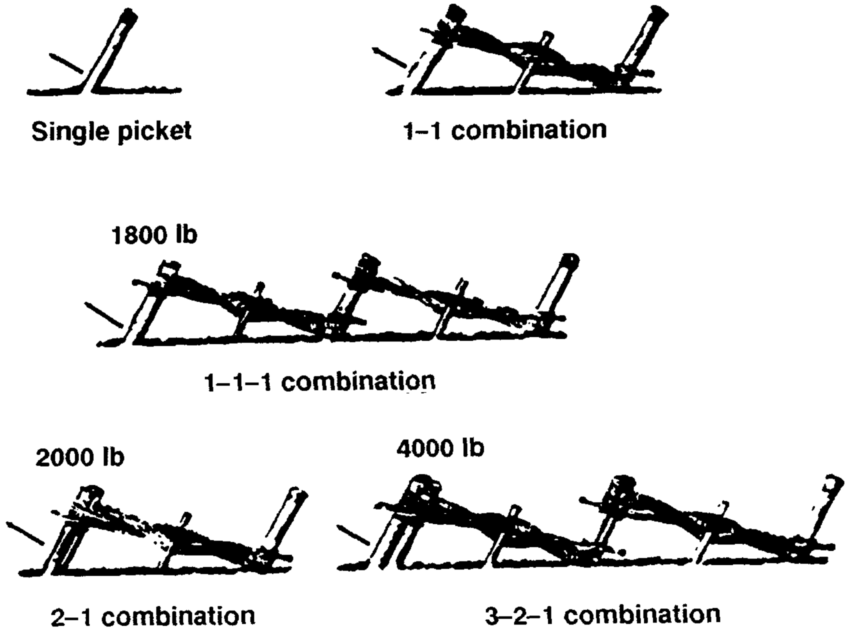


Pickets

Single wooden picket

Wooden stakes used for pickets should be at least 3 inches in diameter and 5 feet long. The picket is driven 3 feet into the ground at an angle of 15 degrees from the vertical and inclined away from the direction of pull.

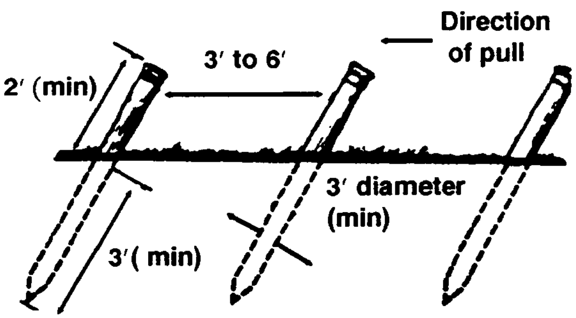
Strengths of picket holdfasts in loamy soil



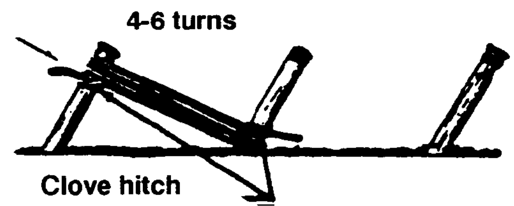
Multiple wooden pickets

The strength of a holdfast can be increased by increasing the area of the picket bearing against the ground. Two or more pickets driven into the ground, spaced 3 to 6 feet apart and lashed together to distribute the load, are much stronger than a single picket. To construct the lashing, a clove hitch is tied to the top of the first picket with four to six turns around the first and second pickets, leading from the top of the first picket to the bottom of the second. Then the rope is fastened to the second picket with a clove hitch just above the turns. To tighten the rope place a stake along the rope and twist it. Then drive the stake into the ground. This distributes the load between the pickets. If more than two pickets are used a similar lashing is made between the second and the third pickets. If wire rope is used for lashing, only two complete turns are made around each pair of pickets. If neither fiber rope nor wire rope is available for lashing, boards may be placed from the top of the front picket to the bottom of the second picket and nailed onto each picket. As pickets are placed farther away from the front picket, the load to the rear pickets is distributed more unevenly. Thus, the principal strength of the multiple picket holdfast can be increased by using two or more pickets to form the front group. This increases both the bearing surface against the soil and the breaking strength.

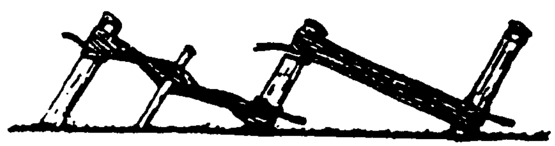
Preparation of a picket holdfast



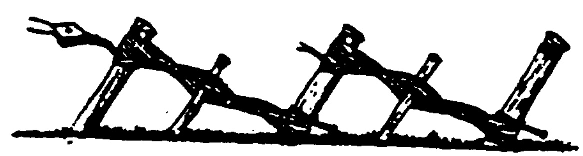
1. Drive pickets (steel or wood) into ground 15″ minimum from vertical



2. Lash pickets together, starting at top of first picket



3. Twist rope with rack stick, then drive stick into ground

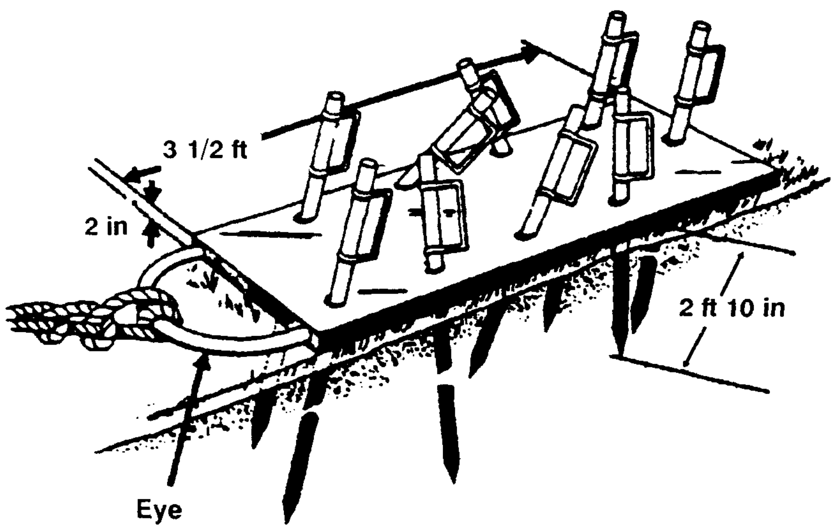


4. Completed picket holdfast

Steel picket holdfasts

A standard steel picket holdfast consists of a steel box plate with nine holes drilled through it and a steel eye welded on the end for attaching a guy line. The pickets are also steel, and are driven through the holes in a way that clinches the pickets in the ground. This holdfast is especially good for anchoring horizontal lines, such as the shore guy on a ponton bridge. Two or more of these units can be used in combination to provide a stronger anchorage. A similar holdfast can be improvised with a chain by driving steel pickets through the chain links in a crisscross pattern. The rear pickets are driven in first to secure the end of the chain and the successive pickets are installed so that there is no slack in the chain between the pickets. A lashed steel picket holdfast consists of steel pickets lashed together with wire rope in the same way as wooden stake picket holdfasts. As an expedient, any miscellaneous light steel members can be driven into the ground and lashed together with wire rope to form an anchorage.

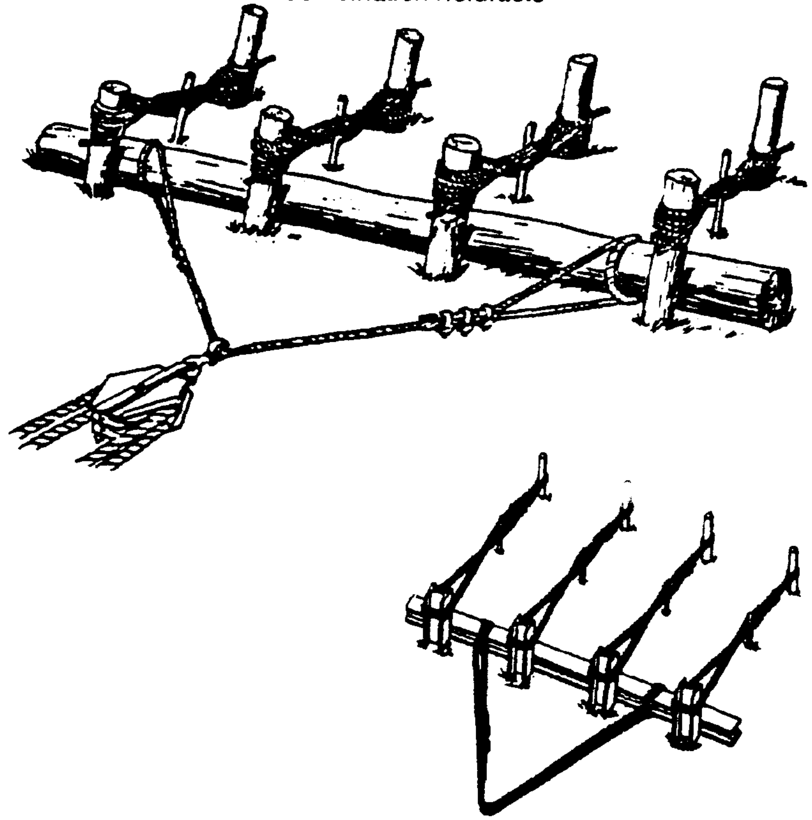
Standard steel picket holdfast



Combination holdfasts

For heavy loading of an anchorage cable, it is desirable to spread the load over the largest possible area of ground. This can be done by increasing the number of pickets used. Four or five multiple picket holdfasts can be placed parallel to each other with a heavy log resting against the front pickets to forma combination log and picket holdfast. The guy line or anchor sling is fastened to the log which bears against the pickets. The log should bear evenly against all pickets in order to obtain maximum strength. The timber should be carefully selected to withstand the maximum pull on the line without appreciable bending. A steel cross member can be used to form a combination steel picket holdfast.

Combination holdfasts

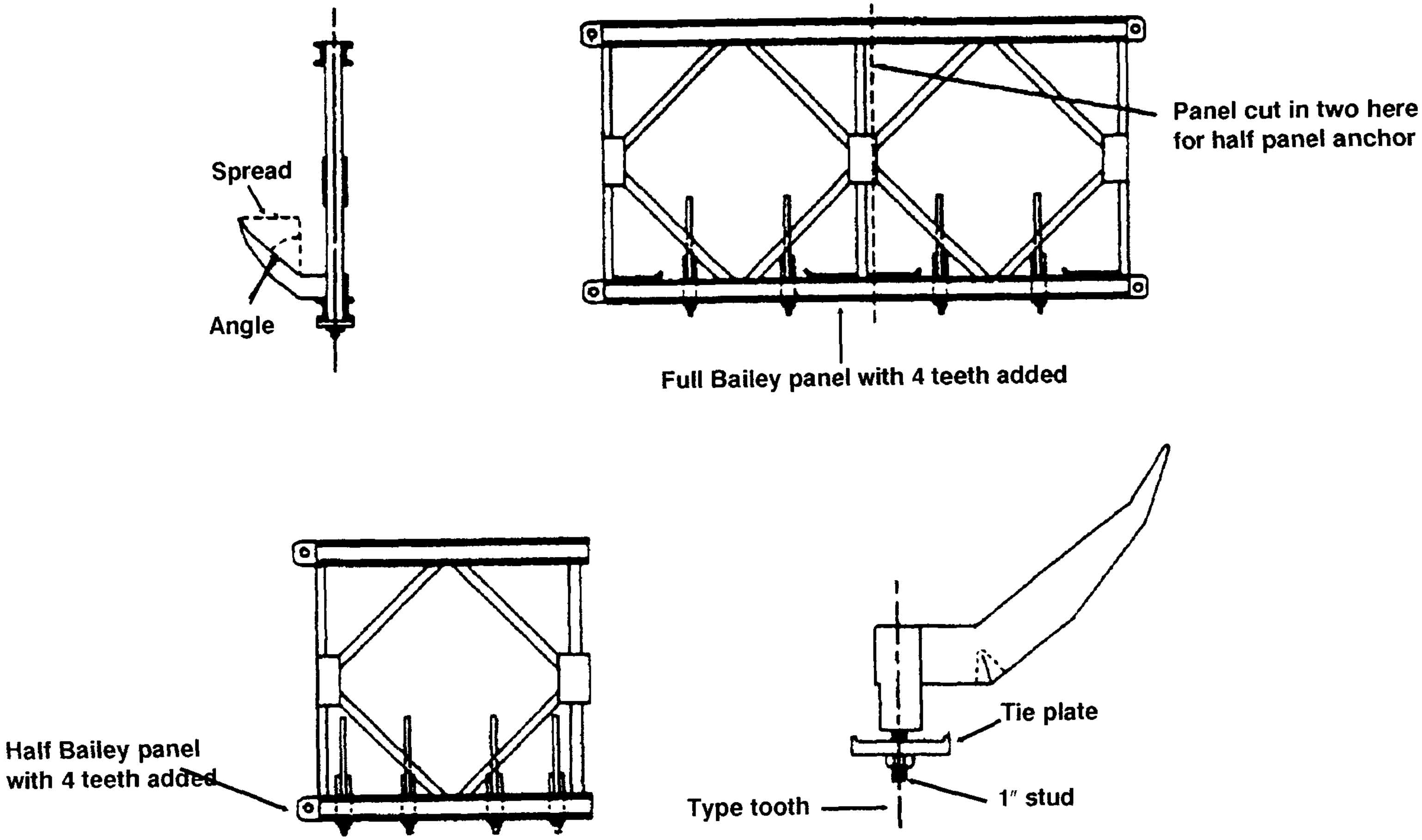


Expedient Anchors

Bailey panel

Bailey or similar bridge panels with teeth attached can provide greater anchor holding capacity than the standard 1OO-pound kedge anchor. The teeth on the Bailey anchor are attached to one chord and the anchor line is attached to the other chord. A half panel anchor provides consistent holding power and is easier to handle than a full panel anchor. The Bailey half panel anchor is recommended when a panel anchor is needed. The high tensile steel teeth used on the panel should be made in the shape of a kedge anchor fluke. Care must be taken when casting this anchor as it has a tendency to turn upside down.

Standard Bailey panel anchors

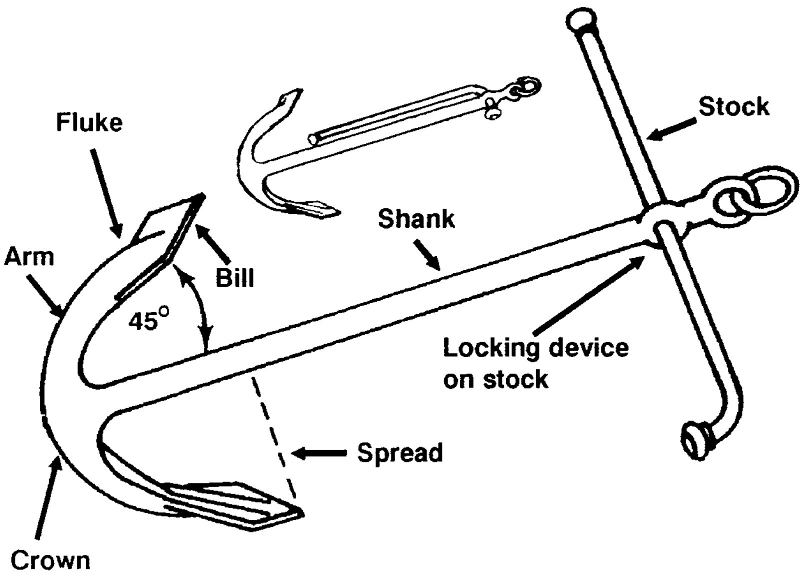


Nonstandard kedge

The following rules should serve as a guide when it is necessary to use other than the standard 100-pound kedge anchor:

- Given the same size and shape, additional weight adds very little to the holding power of an anchor.
- For general use in a wide range of materials, a fluke angle of about 45 degrees gives the greatest holding power. Angle variations, as little as 10 degrees, can result in a change in holding power of 50 percent or more.
- The holding power of anchors of the same size varies with the area of the fluke.

Components of an anchor



Construction of Expedient Anchor Towers Using Bailey Bridge Panels.

Bailey bridge panels with either crib or expedient footings are excellent for use as an anchor tower. Nonstandard parts used in this type tower can be fabricated in the field. Bailey panels can be used to construct anchor towers of heights up to 70 feet. With high towers, horizontal panels are normally added to the base to give greater stability. The panel sections are secured together by field fabricated U-type bolts which are fitted around the panel corners. A crown is made by welding 1 1/2 inch by 1 1/2 inch angle irons to the top of the tower.

Bailey anchor tower